Vijay Sundar Nagamalla

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Forward Deployed Engineer (FDE) | AI Application Developer | Full Stack Engineer

PROFESSIONAL SUMMARY

Technical builder and L&D professional with 8+ years of experience delivering high-scale technical training programs and managing learning operations. Mentored 50,000+ learners in core programming (C, Java, Python, SQL, DSA) across 40+ institutions while operating LMS ecosystems. Proven track record architecting production platforms (React 19, FastAPI, Docker, C#) and intelligent automation tools, targeting Forward Deployed Engineering (FDE) and applied AI solutions.

TECHNICAL SKILLS

AI / ML: RAG Pipelines, MCP (Model Context Protocol), Claude API, OpenAI API, NVIDIA NIM, Whisper ONNX, DeepSeek

Frontend: React 19, TanStack Start, TypeScript, TailwindCSS, Framer Motion, TanStack Query, shadcn/ui

Backend: FastAPI (Python), Node.js, C# / .NET 8 WPF, REST APIs, JWT, RBAC, Judge0

Databases: PostgreSQL, Redis, Supabase, SQLite

DevOps: Docker, Nginx, Vercel, GitHub Pages, GitHub Actions

Languages: Python, JavaScript, TypeScript, C#, Java, SQL

Training and Ops: Technical Training, Power BI, Google Sheets API, MS Excel, LMS, Learning KPIs

PROFESSIONAL EXPERIENCE

Senior Technical Trainer & Learning Operations / AI Developer May 2020 – Present

Freelance

- Architected and deployed 7+ end-to-end AI and full-stack platforms, including multi-platform EdTech analytics (Almaanac, Vukti), open-source AI meeting transcribers (ILFU), and presenter copilots (Forward)
- Delivered high-impact technical training in Java, Python, SQL, and DSA to 25,000+ learners, achieving ~50% average assessment improvement
- Managed end-to-end learning operations: from multi-track session scheduling and batch planning to learner enrollment and attendance tracking
- Partnered with instructional designers and stakeholders to ensure seamless program execution and on-time delivery of learning modules
- Leveraged MS Excel and Power BI to track real-time KPIs, monitor learner completion rates, and maintain operational data with 95%+ accuracy

Technical Project Manager May 2023 – December 2024

Xplore.co.in

- Spearheaded large-scale upskilling programs as Lead Trainer and Project Owner, ensuring 100% on-time delivery of technical training and labs
- Built and maintained interactive Power BI dashboards for leadership to monitor KPIs, learner engagement, and program health in real-time
- Orchestrated end-to-end program operations, including batch planning, stakeholder coordination, and multi-track logistics
- Streamlined training workflows and collaborated with instructional designers to optimize content delivery and assessment results

Associate Trainer June 2018 – May 2020

FACE

- Facilitated high-impact technical training in Aptitude, C, Java, Python, and SQL for batches of 80+ learners
- Developed comprehensive lesson plans, presentations, and hands-on assessments to drive technical proficiency and student outcomes
- Managed end-to-end classroom operations, from attendance tracking to performance evaluation and progress reporting
- Provided structured feedback and personalized learning support to enhance student technical skills and interview readiness

KEY PROJECTS

Almaanac — Multi-Platform Classroom Coding Intelligence Live Platform

TanStack Start (React 19), TanStack Router, TanStack Query v5, Supabase (PostgreSQL/RLS), Tailwind CSS v4, Recharts, TypeScript 5.8, Vercel

- Classroom analytics platform aggregating coding metrics across 8 platforms: LeetCode, Codeforces, AtCoder, GeeksforGeeks, HackerRank, CodeChef, HackerEarth, and Code360
- Engineered independent async refresh workers with rate-limiting, lease-based Postgres locking, and failure isolation per platform
- Delivered educator dashboards: difficulty-weighted scoring, leaderboards, behavioral cohorts ("Active Today", "At Risk"), 365-day heatmaps, and term-end PDF report rollups

Vukti — Unified Academic Intelligence Platform & Student OS Live Platform

React 19.2, Vite 7, TailwindCSS 4, FastAPI, SQLAlchemy 2, PostgreSQL 16, Redis 7, Docker, Judge0 1.13.1, Monaco Editor

- Architected and deployed a 9-module student OS for technical institutions (B2B + B2C) across Academics, Placements, and Growth for 10,000 concurrent users
- Multi-platform performance sync across 6 platforms (LeetCode, HackerRank, CodeChef, Codeforces, GFG, HackerEarth) with async worker pool
- Full examination engine: MCQ + Coding + SQL testing with Docker-isolated Judge0 1.13.1 (25+ languages), SQL judge, and proctoring/tab-blur detection
- VIMY gamified competitive coding: 1,000+ problems, Monaco editor, streaks, coins, 30 XP levels, leagues, heatmaps, and 12-domain visual Skill DNA

Salesforce Tracker — Trailhead Profile Scraper Live Platform

TanStack Start (React 19), TanStack Router, Supabase, Tailwind CSS v4, shadcn/ui, XLSX, TypeScript

- Bulk-scrape Salesforce Trailhead profiles from URLs or Excel uploads for institutional tracking and certification monitoring
- Auto-detect Trailhead profile columns, batch-scrape with progress tracking, and merge with uploaded metadata
- Persistent session history with export to Excel and server-side scraping via TanStack Start API routes

FDE Flow Atlas — Interactive Learning Platform Live Platform

Vanilla HTML/CSS/JS, SVG Animations, LocalStorage Progress, GitHub Pages

- 133-topic interactive curriculum for Forward Deployed Engineering with animated SVG flow diagrams, case studies, and 11 hands-on projects
- Step-by-step narration with code highlighting, playback controls, interactive quizzes, and fill-in templates
- Covers RAG pipelines, tool calling, autonomous workflows, case decomposition — all progress tracked via localStorage

ILFU (I Listen For U) — Open-Source Local Video Call Transcriber GitHub Repository

Python 3.11+, faster-whisper, PyAudioWPatch (WASAPI), whisper.cpp, Pytest, AGPL-3.0

- Free, open-source meeting transcriber running 100% locally on device with zero API keys, no accounts, and NO bot joining the call
- Engineered passive system audio loopback capture (WASAPI / PyAudioWPatch) to record microphone and speaker audio without call interference
- Integrated faster-whisper & whisper.cpp for on-device STT, open speaker diarization, and local LLM summarization with capped CPU/memory usage

Forward — Non-Intrusive Presenter & Live-Instruction Copilot

C# WPF, .NET 8, Claude API, DeepSeek, NVIDIA NIM, NAudio WASAPI, Whisper ONNX, AMD XDNA NPU

- Built a live-instruction copilot allowing educators and presenters to view real-time AI prompts and live transcription while screen-sharing without audience clutter
- Hardware-level window isolation via WDA_EXCLUDEFROMCAPTURE P/Invoke keeps speaker notes private and invisible to meeting screenshares (Zoom, Teams, Meet)
- Multi-provider AI fallback chain: Claude → DeepSeek → NVIDIA NIM → Groq → Ollama with automatic failover
- Hardware-optimized on-device AI: Whisper STT on AMD XDNA NPU, embeddings on Radeon DirectML GPU, Silero VAD on CPU
- Hotkey-driven UX: Alt+Space (toggle overlay), Ctrl+Space (explain last 30s), Alt+A (focus input)

EDUCATION

Bachelor of Technology (B.Tech) in Electronics and Communication Engineering Graduated 2019

North East Christian University